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EXPERIMENT 2

Aim: Develop Software Requirement Specification (SRS) document in IEEE format for the project.

Software Requirements Specification (SRS) for Agriculture App

Version: 1.0

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Prepared by: Gargi S

1. Introduction

1.1 Purpose

This document defines the functional, non-functional, and system requirements for the agriculture app, which provides **a community-driven platform with chatbot assistance and multilingual support** to help farmers collaborate and make informed decisions.

1.2 Document Conventions

This document follows the IEEE 830-1998 standard for SRS documentation.

1.3 Intended Audience and Reading Suggestions

- **Developers:** For understanding system design and API integration.
- **Project Managers:** To track project scope and deliverables.
- **Stakeholders (Farmers, Agronomists, Agri-businesses):** To ensure user needs are met.

1.4 Product Scope

The app provides **community engagement, AI chatbot support, and multilingual accessibility** for better farming decisions.

2. Overall Description



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2.1 Product Perspective

- **Standalone Mobile/Web Application**
- **Integration with AI Chatbot & Cloud-based Data Processing**

2.2 Product Functions

1. **User Registration & Authentication**
2. **Community Forums & Groups**
3. **AI Chatbot Assistance**
4. **Multilingual Support**
5. **Weather API Integration**
6. **Crop & Soil Health Monitoring**

2.3 User Characteristics

- **Farmers:** Need simple UI & voice support.
- **Agronomists:** Require advanced analytics.
- **Admins:** Manage users & moderate discussions.

2.4 Constraints

- **Internet Dependency**
- **Device Compatibility**
- **Data Privacy Compliance**

3. Specific Requirements



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3.1 Functional Requirements

ID	Requirement	Description
FR-1	User Authentication	Users log in via email/phone, password, or biometric authentication.
FR-2	Community Forum	Users can create, join, and participate in discussions.
FR-3	AI Chatbot	Provides instant farming advice and troubleshooting.
FR-4	Multilingual Support	Users select their preferred language for UI & chatbot responses.
FR-5	Weather API Integration	Fetches and displays real-time weather data.
FR-6	Crop & Soil Health Monitoring	Users input crop details for AI-based recommendations.

3.2 Non-Functional Requirements

- **Performance:** Load time ≤ 3 seconds.
- **Security:** AES-256 encryption for user data.
- **Scalability:** Support [X] concurrent users.

3.3 External Interface Requirements

- **APIs:** Weather API, AI Chatbot API, Translation API.
- **UI/UX:** Must follow **Material Design (Android) / Human Interface Guidelines (iOS)**.

4. System Architecture & Diagrams

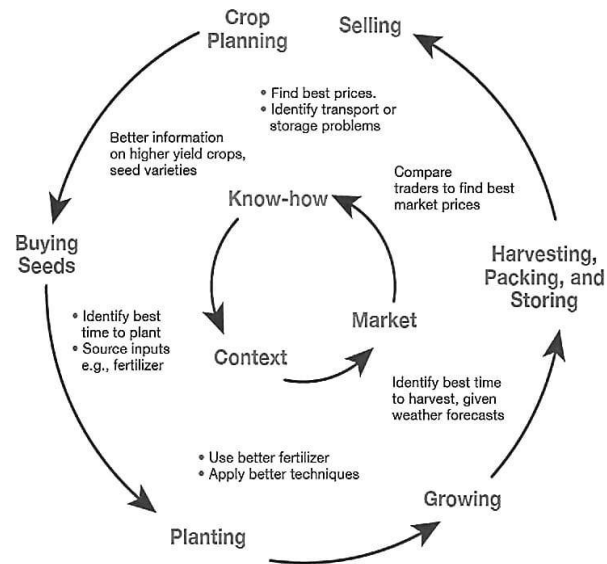
This section provides a visual representation of the system architecture, workflows, and data flow.

4.1 Proposed System

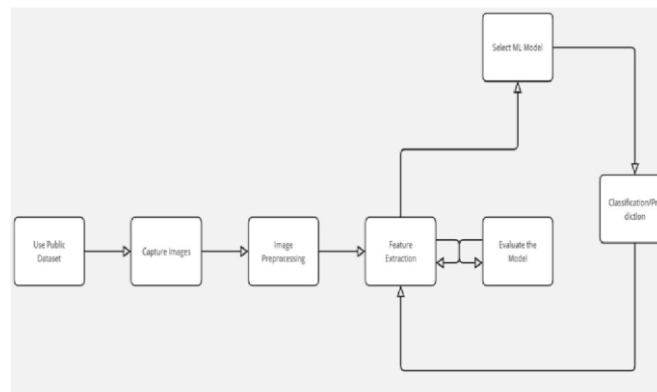


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4.5 Flowchart



5. Appendices

- **References:**

- **Standards & Guidelines:**

- IEEE 830-1998 – Recommended Practice for Software Requirements Specifications.
- ISO/IEC 25010:2011 – Software Quality Model.
- Agile Manifesto – <https://agilemanifesto.org> (For Agile Development).

- **APIs & External Services:**



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- OpenWeather API – <https://openweathermap.org/api> (For real-time weather updates).
- Google Translate API – <https://cloud.google.com/translate> (For multilingual support).
- AI Chatbot API (e.g., OpenAI, Dialogflow) – [Insert API documentation link].

➤ **Frameworks & Technologies Used:**

- React Native / Flutter – Mobile app development frameworks.
- Node.js / Django / Spring Boot – Backend technologies.
- Firebase / PostgreSQL / MongoDB – Database solutions.

• **Glossary:**

Term	Definition
AI Chatbot	An artificial intelligence-based system that provides automated responses and farming assistance.
API (Application Programming Interface)	A set of protocols and tools that allow different software applications to communicate.
Authentication	The process of verifying a user's identity before granting access to the system.
Backend	The server-side logic and database management that handle requests from the frontend.
Community Forum	A feature where users can post discussions, ask questions, and interact with other farmers and agronomists.
Crop & Soil Health Monitoring	The process of analyzing crop conditions and soil health using AI-based recommendations.
DFD (Data Flow Diagram)	A visual representation of how data moves through the system.
Frontend	The part of the application that users interact with, such as mobile or web UI.
Iteration	A cycle in Agile development where features are built, tested, and improved based on feedback.
Multilingual Support	The ability of the app to provide content and chatbot responses in multiple languages.
Sprint	A time-boxed development cycle in Agile methodology where a specific set of features is built and tested.
System Architecture	The design and structure of the software system, including components and their interactions.



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UAT (User Acceptance Testing)	The process of testing the application with actual users to ensure it meets their requirements.
Weather API	An external service that provides real-time weather data to the application.